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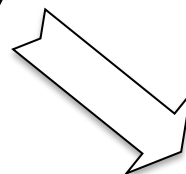
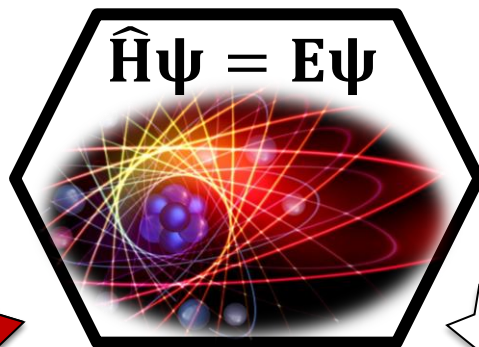
Research Training Group 2721: HYDROGEN Isotopes $^1,^2,^3\text{H}$

Computational modelling of atomistic motifs for adsorptive He isotope separation

Elvira Gouatieu Dongmo

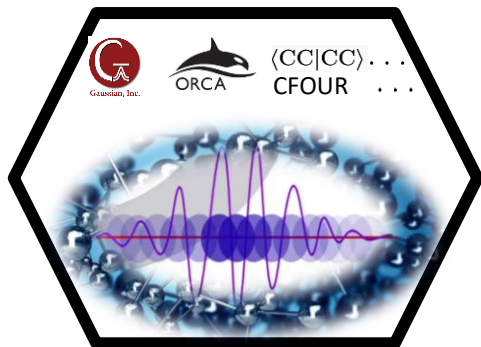
Supervision: Prof. Dr. Thomas Heine

$^1,^2,^3\text{H}$ Colloquium - Leipzig, 28.09.2023



Quantum Chemistry

(Jensen, Frank. John Wiley & Sons, 2017.)



Numerical Methods

(MAZUMDER, Sandip. Academic Press, 2015.)



Machine Learning

(DRAL, Pavlo O. QC in Elsevier, 2022.)

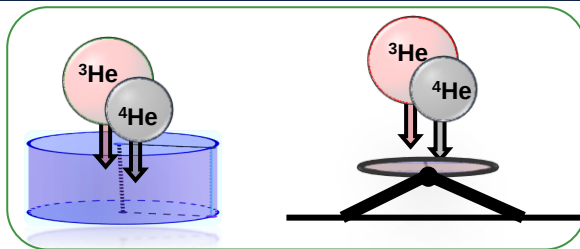


Accurate 
approximations

Computational time 

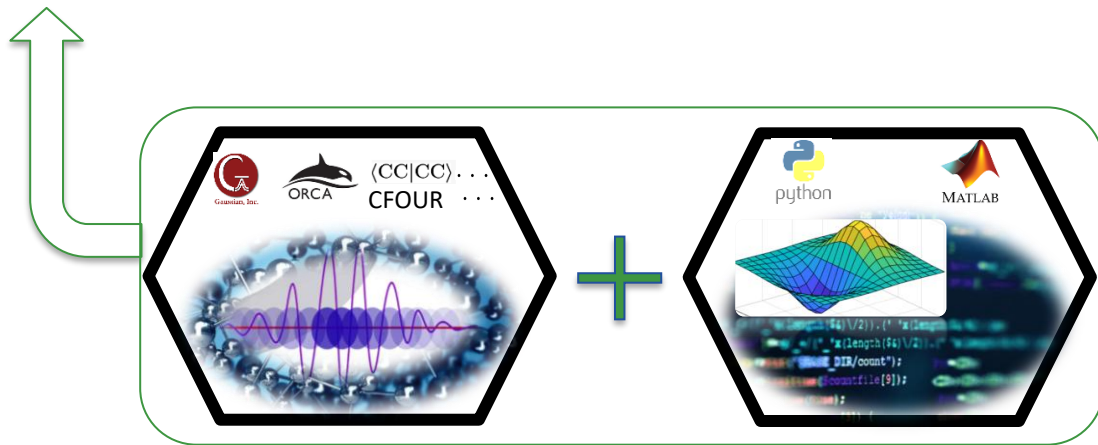
Idea

Helium isotope separation



Nuclear quantum effect (NQE)

Zero point energy (ZPE)

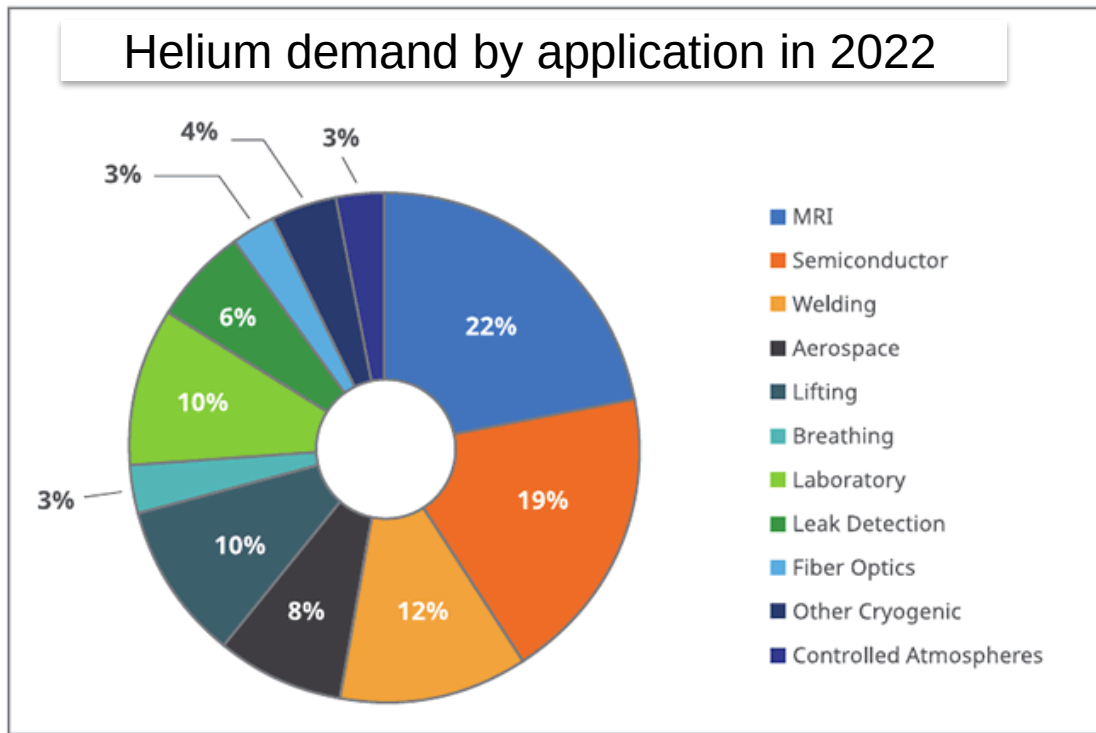


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H₂ isotopologue separation

Why Helium?



Industrial separation techniques



name	Helium-4	Helium-3
symbol	^4He	^3He
mass	4.00260324 u	3.01604927 u
Natural abundance	99.99987%	0.00013%
half-life ($t_{1/2}$)	stable	stable

Separation techniques



Cryogenic distillation

Pressure-swing adsorption

Membrane separation from natural gas



Very expensive

Low temperature

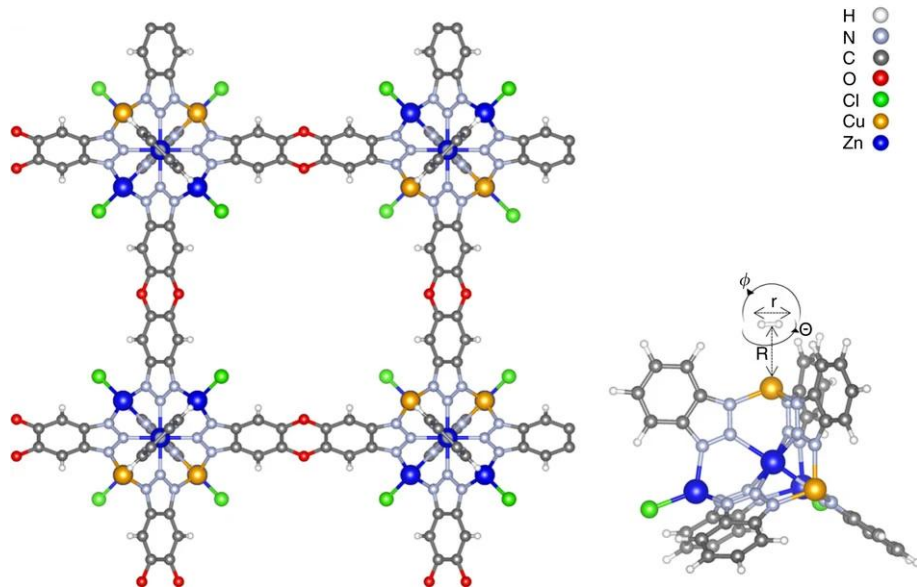


Alternative approaches



Hydrogen isotopologue separation

The MOF Cu(I)-MFU-4l



Used **CAQS** to increase the hydrogen isotope separation strength.

$\Delta_{\text{ad}}H(\text{H}_2) = -32.7 \text{ kJ}\cdot\text{mol}^{-1}$
 $\Delta_{\text{ad}}H(\text{D}_2) = -35.0 \text{ kJ}\cdot\text{mol}^{-1}$

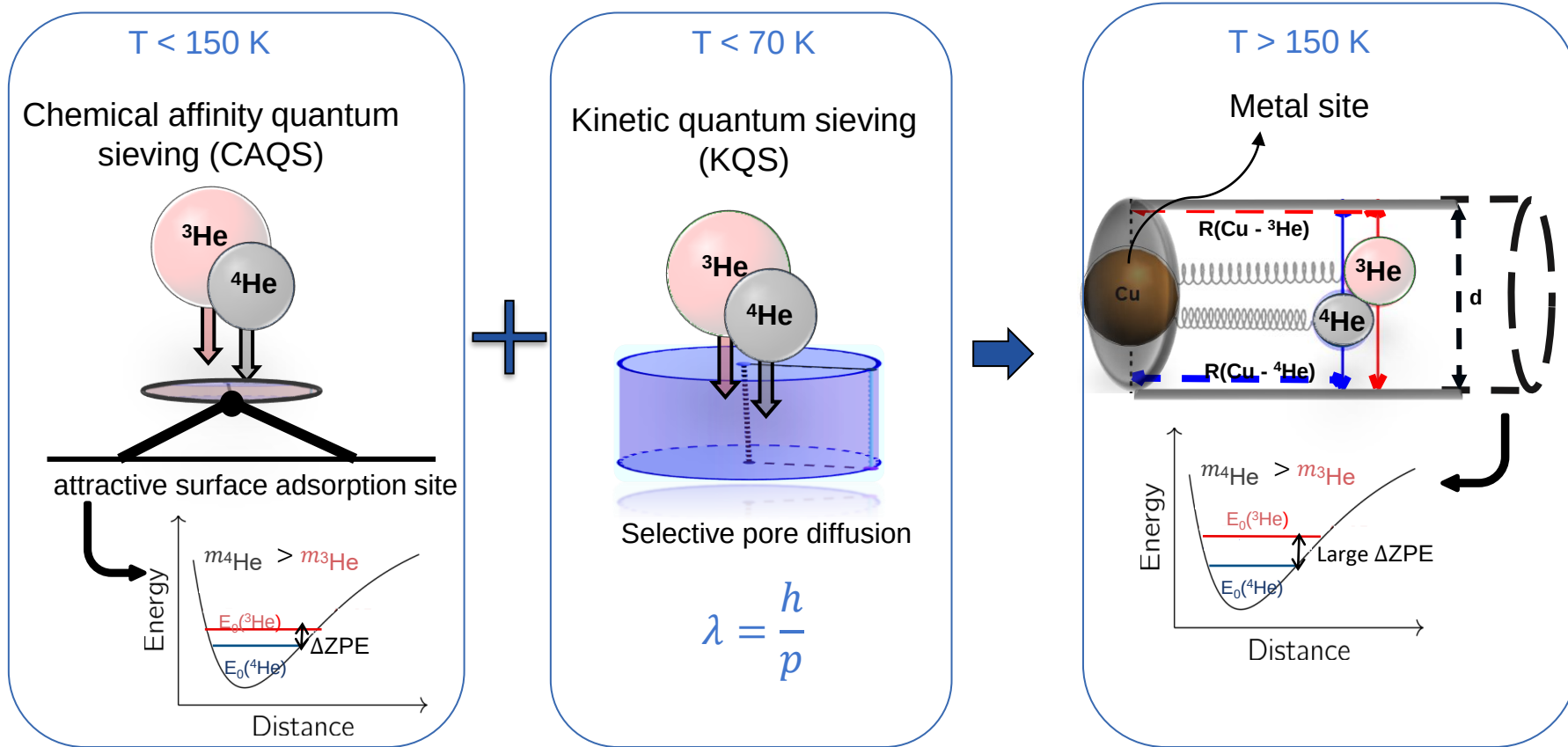
difference in adsorption
enthalpy: $\Delta_{\text{ad}}E = -2.3 \text{ kJ}\cdot\text{mol}^{-1}$

Selectivity: $\text{D}_2\text{-over-H}_2 = 11$ ($T = 100 \text{ K}$)

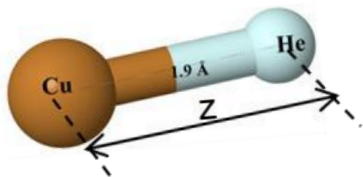
CAQS = Chemical affinity quantum sieving

How about using
CAQS to separate
He isotope? 🤔

Research aim



Model system and Methods

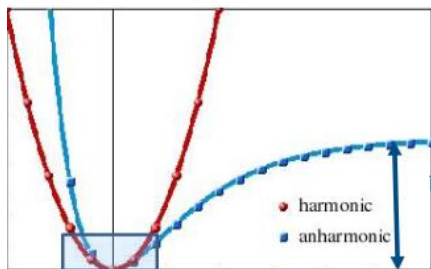


👉 **Analytical calculations :** Separation of variables and zeros of Bessels functions

👉 **Numerical approaches :** **ab initio calculations** and **finite difference method (FDM)**

✓ VPT2 (beyond harmonic oscillator)

✓ CCSD(T)/aug-cc-pVTZ



VPT2= Second-order vibrational perturbation theory

✓ $\hat{H}\Psi = E\Psi$

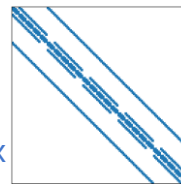
✓ $\hat{H} = T + V = -\frac{\hbar^2}{2m^*} \nabla^2 + V_C(r) + V_M(z)$

Kinetic energy Potential energy

✓ $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

in 3D

5x5x5 square
Hamiltonian matrix



Potential energy

✓ $\hat{H}\Psi = E\Psi$

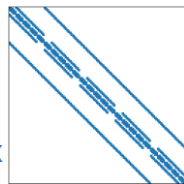
✓ $\hat{H} = T + V = -\frac{\hbar^2}{2m^*} \nabla^2 + V_M(r) + V_C(z)$

Kinetic energy

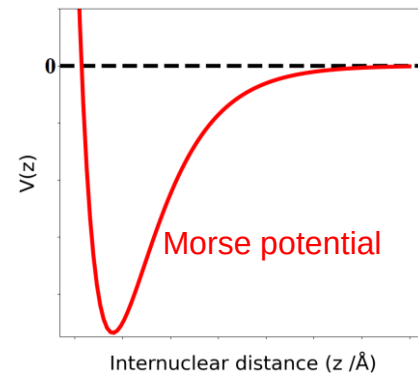
Potential energy

✓ $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

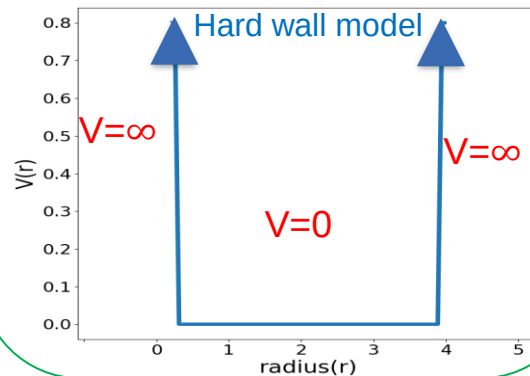
in 3D



5x5x5 square Hamiltonian matrix



+

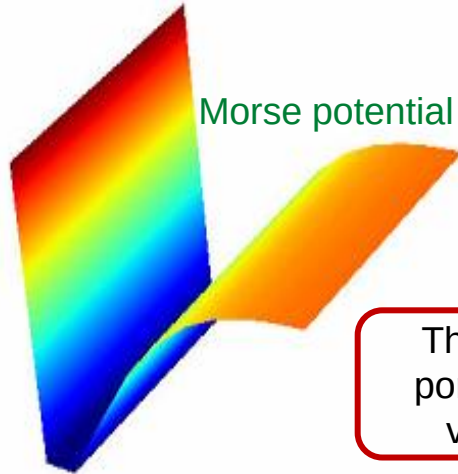


2D Potential energy profiles

Range in Å :

$[-1.5, 1.5]$ along the r-axis

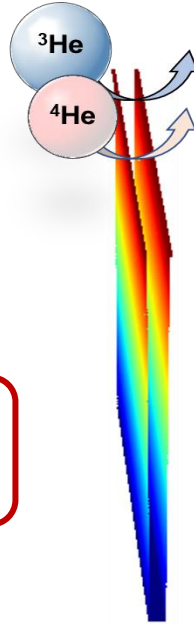
$[1.5, 4.5]$ along the z-axis



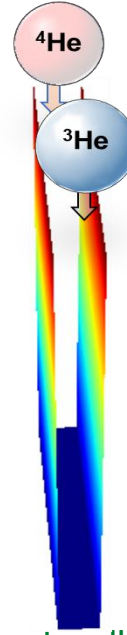
$N = 100$ points; step = 0.03

The choice of the pore size radius is very important

✓ The pore size significantly reduced: **no penetration**

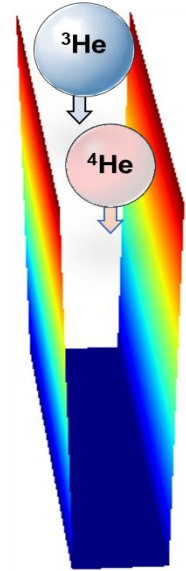


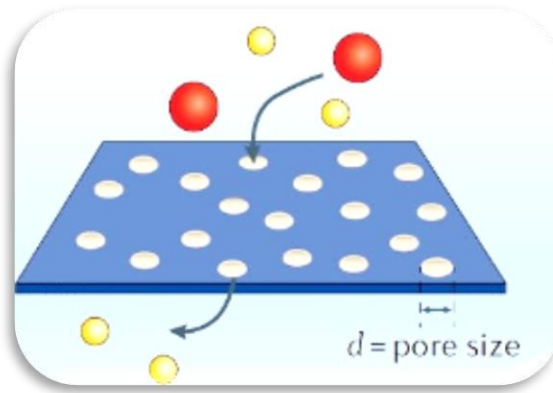
✓ The pore size is smaller enough: **He isotope are adsorbed**



Infinite hard walls

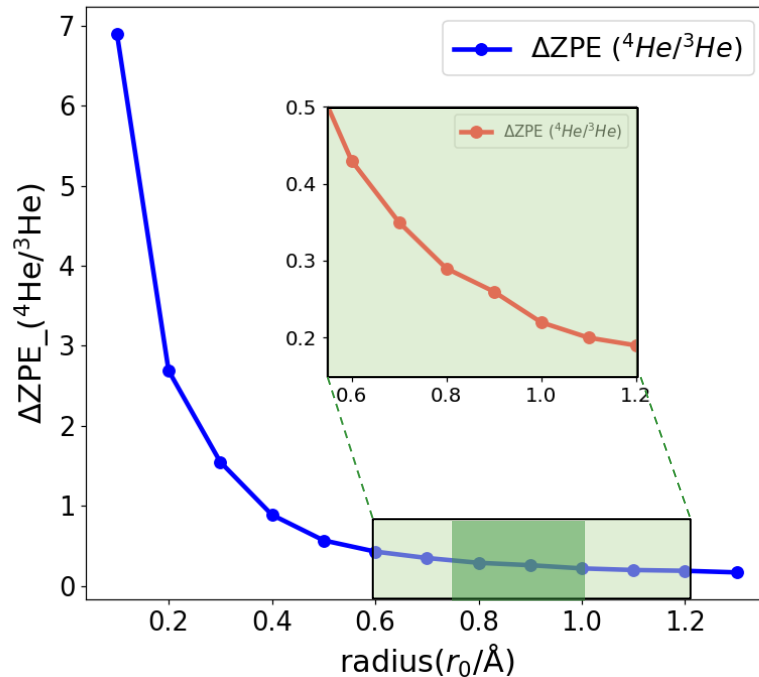
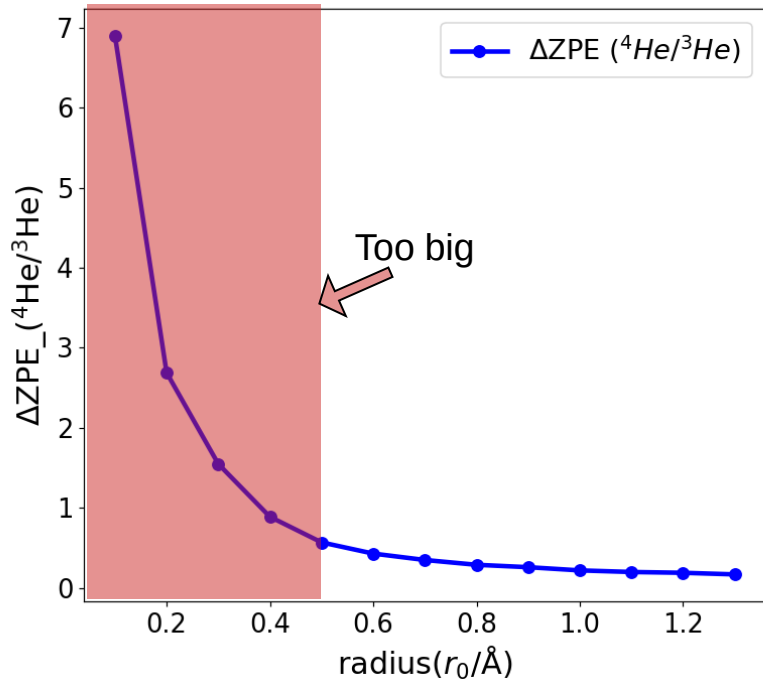
✓ Very large pore size: **minimal impact of the confining potential**



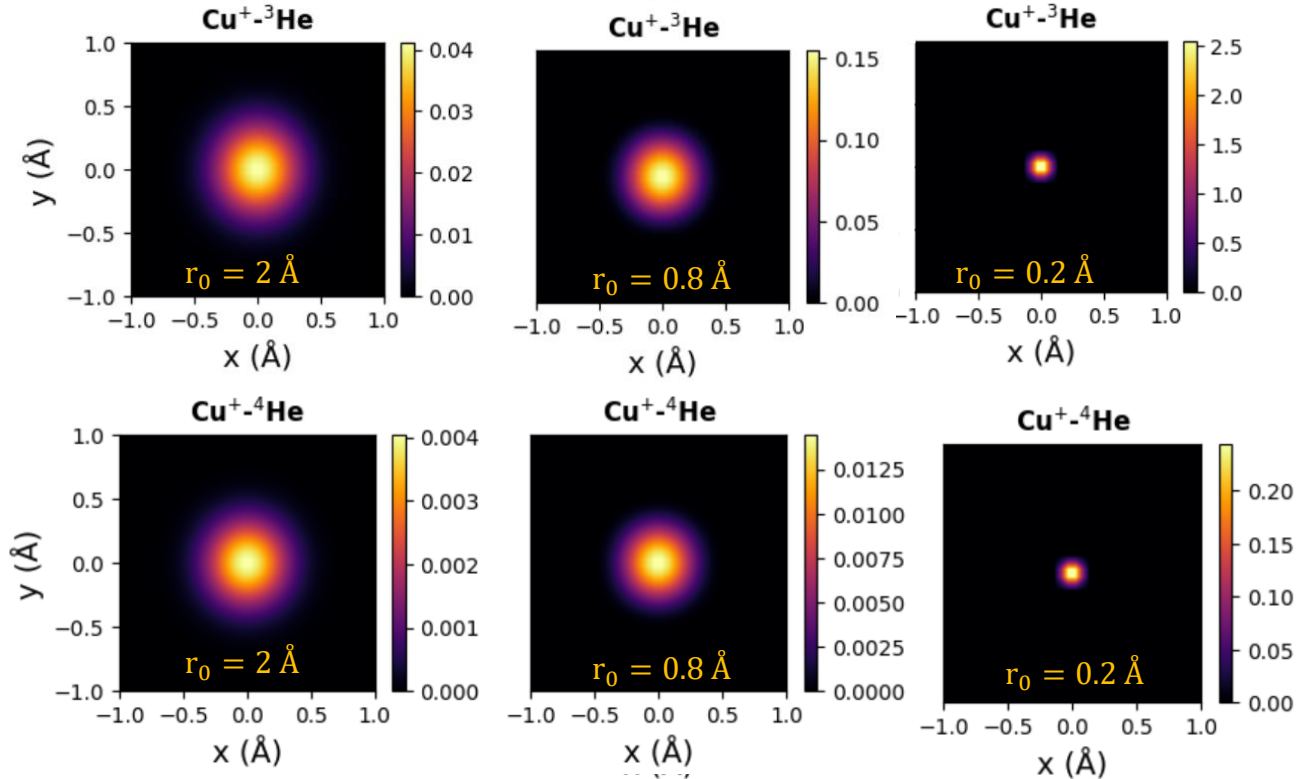


How small is small enough for quantum confinement?

Effect of confinement on ΔZPE



Effect of confinement on the wavefunctions



No confinement

Confined space

$[-1.5, 1.5] \rightarrow r\text{-axis}$

$[1.5, 4.5] \rightarrow z\text{-axis}$

$N = 35$ points

Δr (step) = 0.02

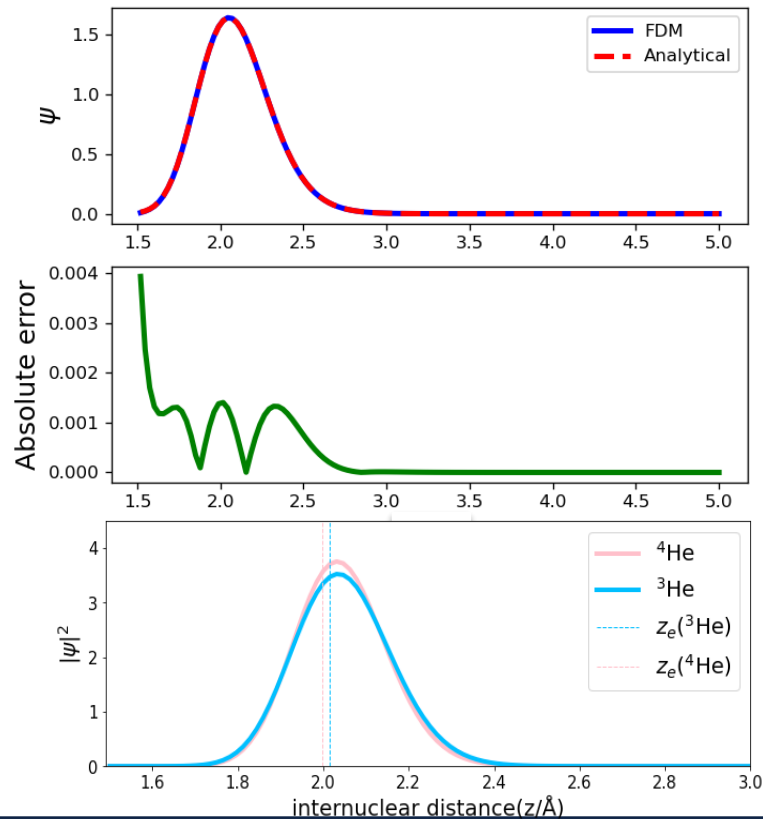
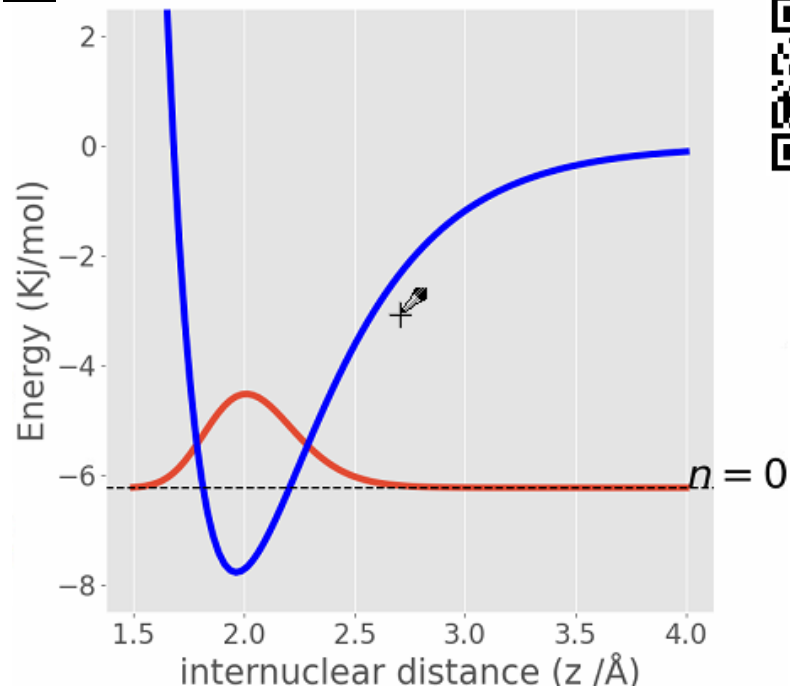
Δz (step) = 0.02



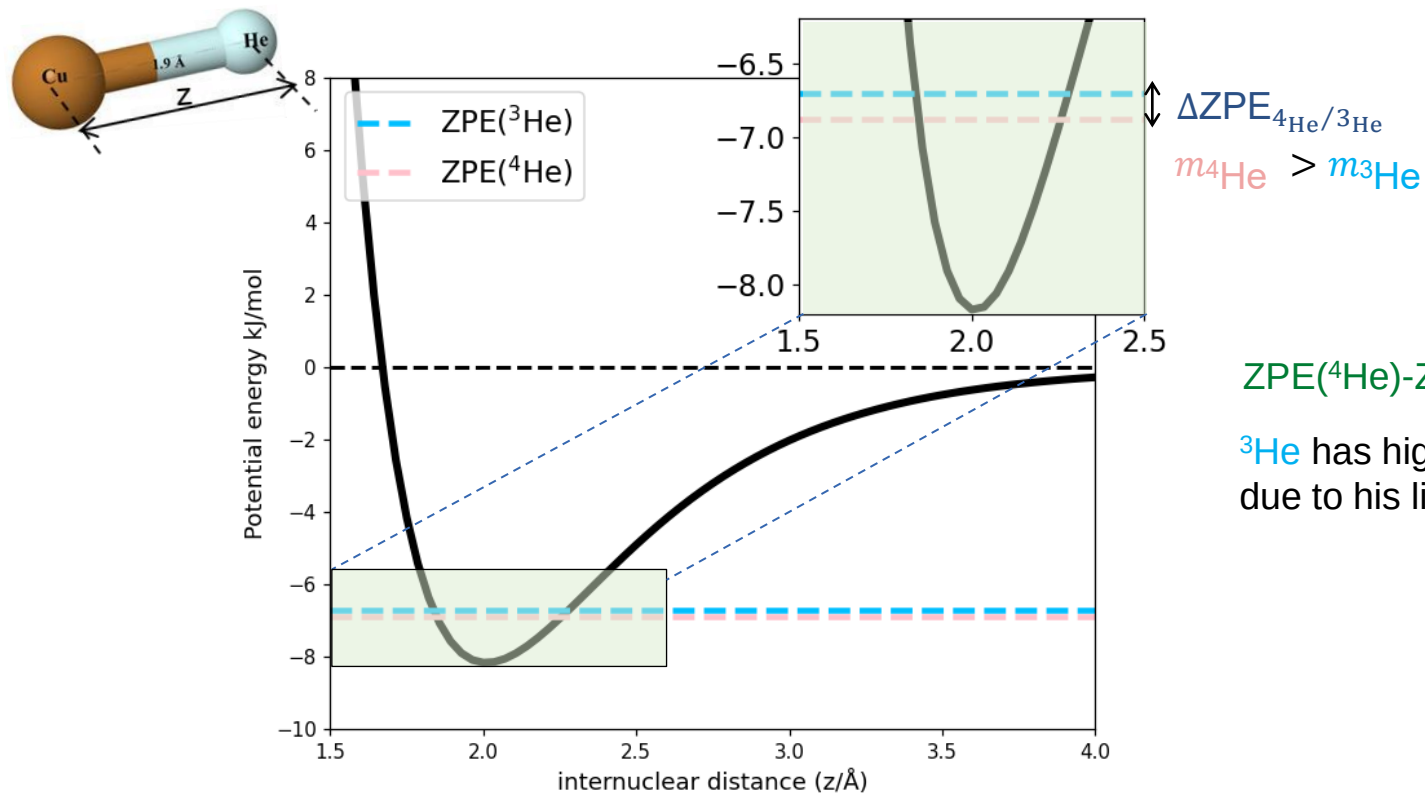
Anharmonic ZPE: Numerical outcome for He isotope

Validation and Quantum states

https://github.com/elvira123hub/Animation_1D_SE_Morse/



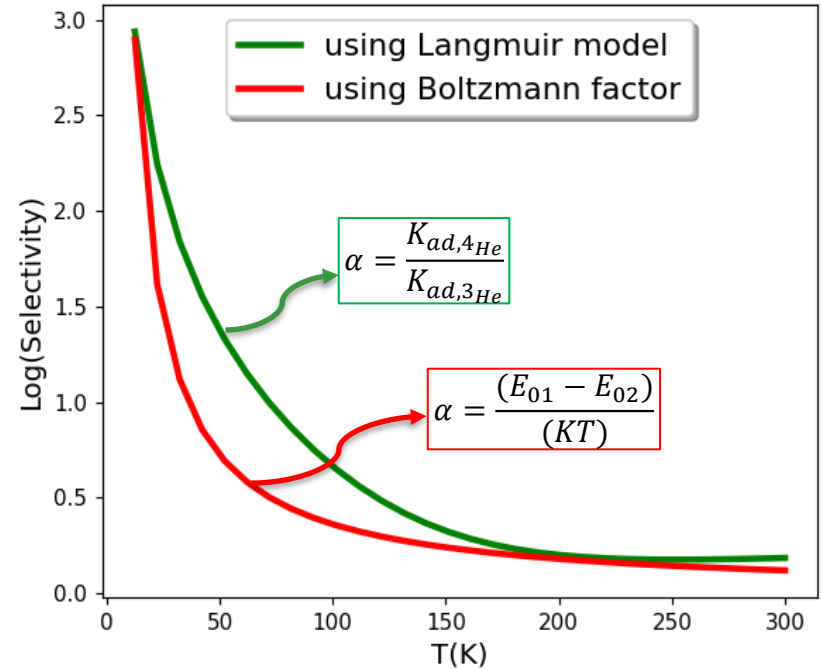
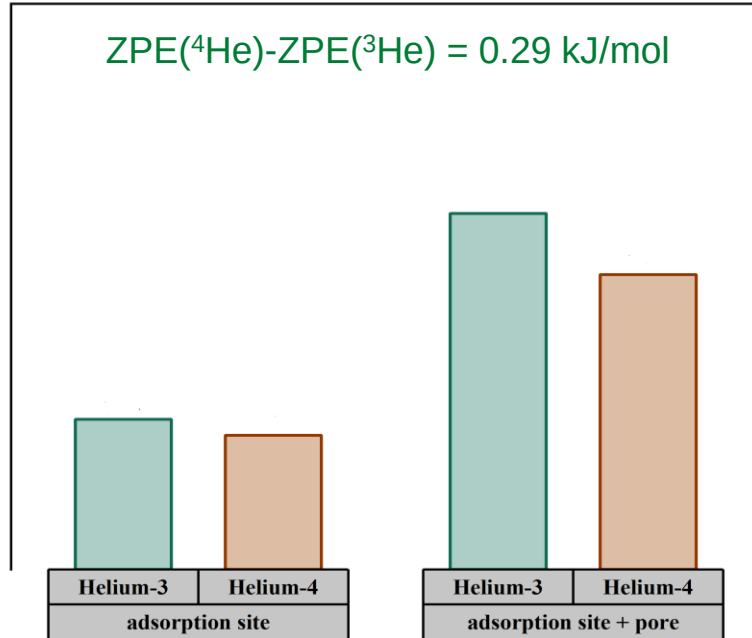
Zero point energy of helium isotope



$\text{ZPE}(^4\text{He}) - \text{ZPE}(^3\text{He}) = 0.17 \text{ kJ/mol}$

^3He has higher ZPE compare to ^4He due to his lighter nuclear mass.

Enhanced ΔZPE



Computational time

Methods	ΔZPE kJ/mol	Time
Analytic	0.171	/
FDM ^w	0.172	0.05 s
VPT2	0.180	54 min
FDM ^c	0.301	23 min

^wZPE considering only the Morse potential.

^cZPE with presence of an infinite hard wall model ($r_0 = 0.8 \text{ \AA}$) in a 3D Scheme.

Summary

$\Delta\text{ZPE}({}^4\text{He}/{}^3\text{He})$ enhanced?



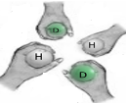
The $\Delta\text{ZPE}({}^4\text{He}/{}^3\text{He})$ **INCREASE** from **0.17 kJ/mol** to **0.29 kJ/mol**.

Temperature requires for the selectivity



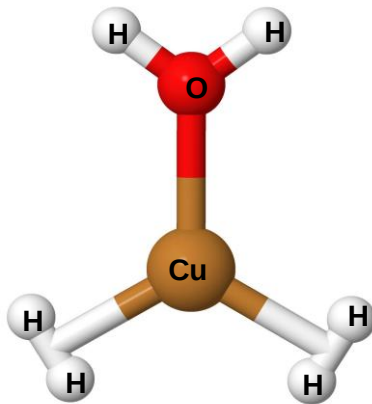
The selectivity of He adsorption is dominated by the ZPE, $\alpha=2.6$ at only **T=77K**

Hydrogen Isotopologues

Collab  ration



Prof. Daniel Boese



Ongoing paper



Shabnam Haque

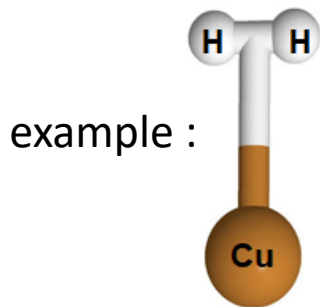
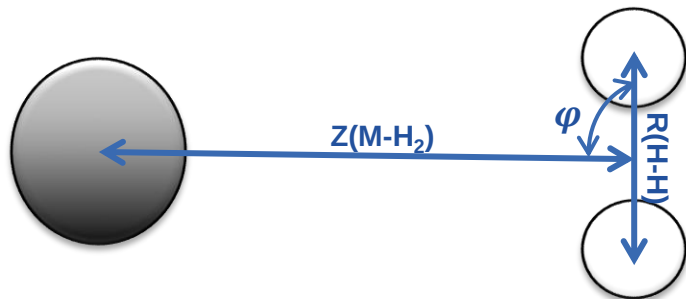


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Strong anharmonicity in the vibrational spectra of $\text{Cu}^+(\text{H}_2\text{O})(\text{H}_2/\text{D}_2)_n$ isotopologues

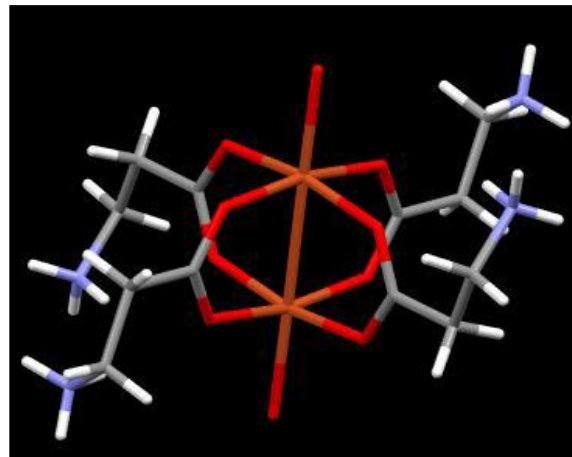
Future work: Hydrogen Isotopologues

FDM + ab initio calculations



And MOFs

CuPW-Ala



Formula: $[\text{Cu}_2(\text{H}_2\text{O})_2(\text{C}_3\text{H}_7\text{NO}_2)_4](\text{NO}_3)_4 \cdot 4\text{H}_2\text{O}$
Molecular weight: 839.6 g/mol

Metal ion: **Copper**
Ligand: **β -Alanine**
Anion: **Nitrate**
Solvent: **Water**

THANK YOU!

👉 Prof. Dr. Thomas Heine

👉 Dr. Toshiki Wulf

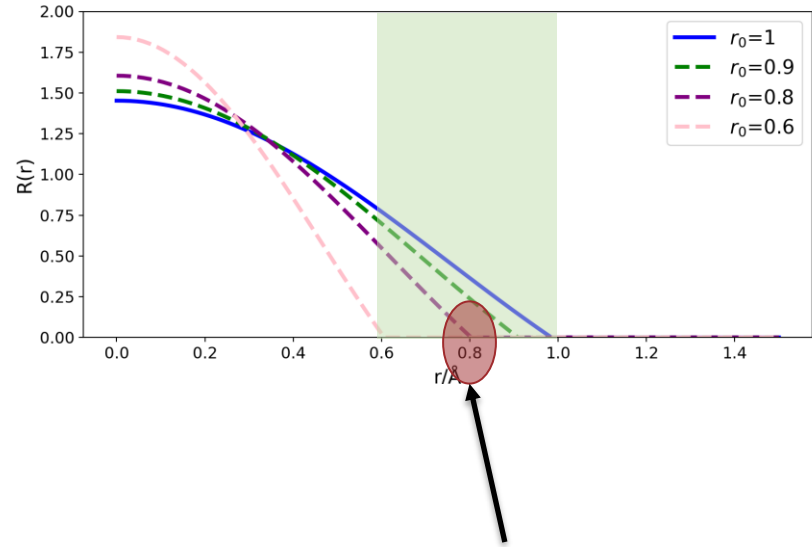
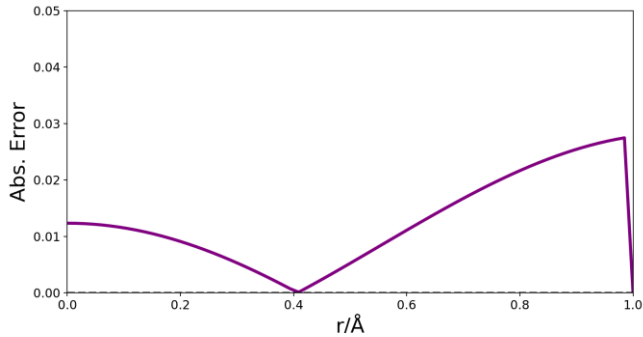
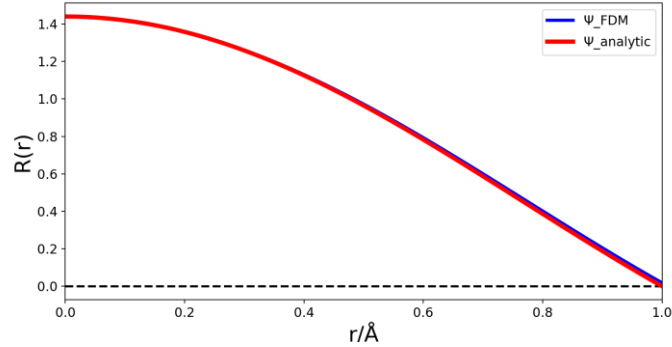
👉 Moritz Leucke



backup slides

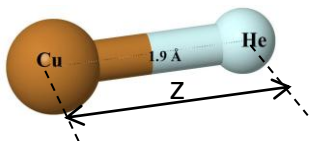


Effect of confinement on the wavefunctions



By obtaining a reasonable value for a pore confinement radius with less error.

Interaction potential energy



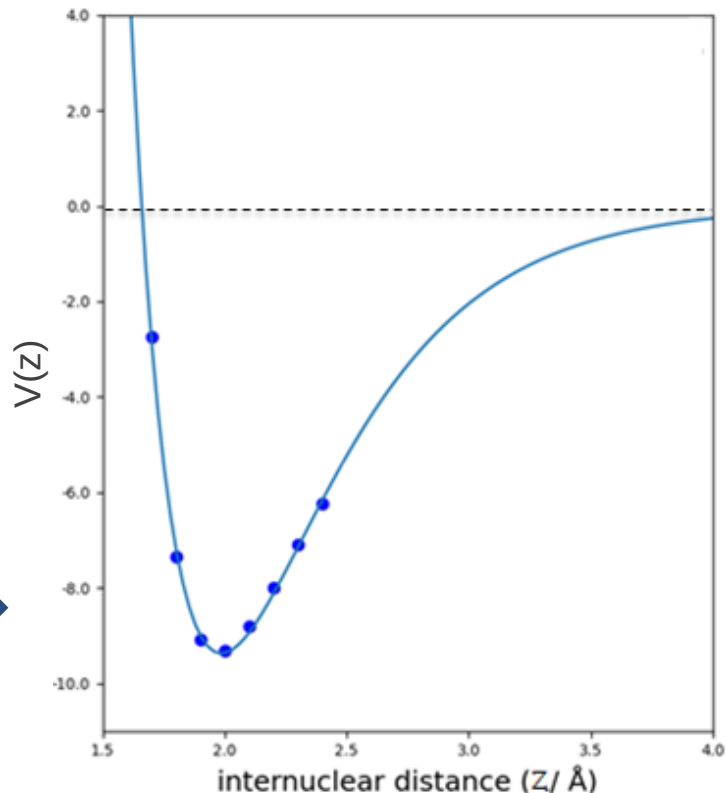
❖ Good approximation: **Morse potential**

$$V(z) = D_e \left[\left(1 - \exp(-\alpha(z - z_e)) \right)^2 - 1 \right]$$

D_e : dissociation energy, z_e : bond length, α (curvature)

Fitted Morse potential at the
CCSD(T)/aug-cc-pVTZ level of theory

$$D_e = 8.17 \text{ kJ} \cdot \text{mol}^{-1} \quad \alpha = 2.04 \text{ \AA}^{-1} \quad z_e = 2.01 \text{ \AA}$$



Numerical solution

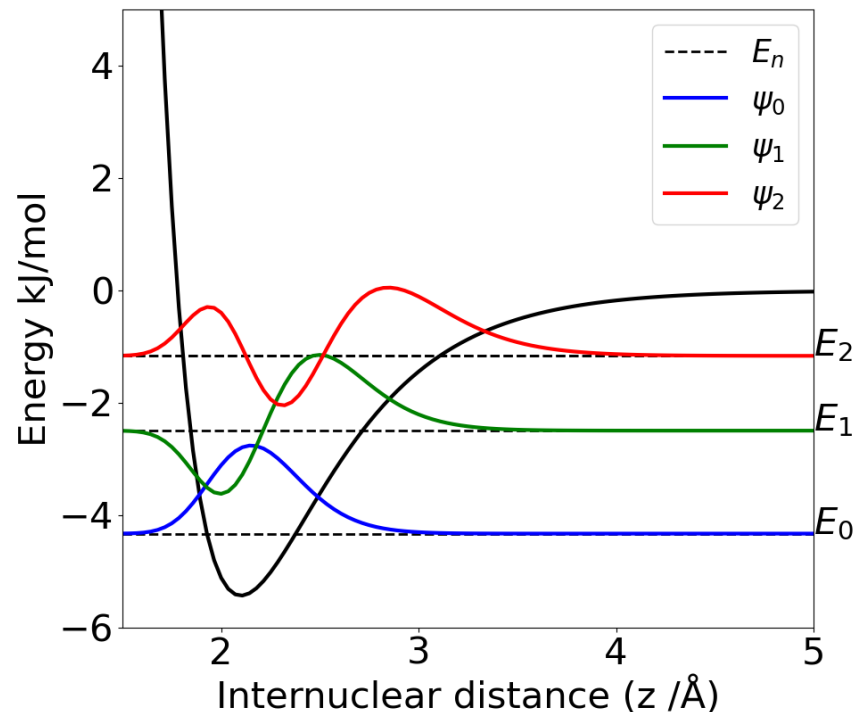
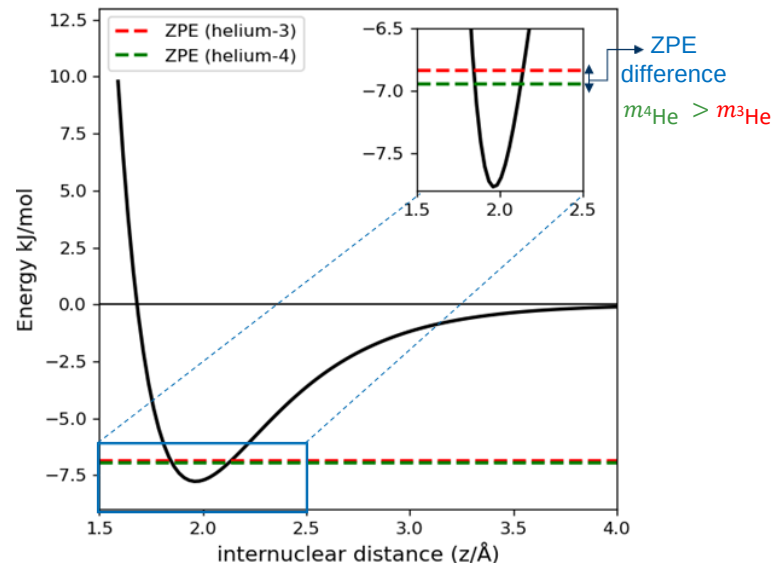


Fig. 1 : Numerical solution of the Schrödinger equation with the Morse potential at the CCSD(T)/cc-pVTZ level.

□ ZPE difference



□ Accuracy and time complexity

Grid points (N)	Error of approximation	Computational time (secs)
100	0.0032	0.044
150	0.0014	0.054

Comparison

Table 1: Short analysis of the results : At the CCSD(T)/cc-pVTZ level of theory with the Morse parameters ($D_e=5.43 \text{ kJ. mol}^{-1}$, $\alpha = 2.16 \text{ \AA}^{-1}$, $z_e=2.1 \text{ \AA}$)

Energy	Numerical Results (kJ. mol ⁻¹)	Analytical results (FDM)	Relative error
E_0	-4.3265	-4.3261	0.0093
E_1	-2.4956	-2.4941	0.0601
E_2	-1.1653	-1.1635	0.1545

Analytical solution

Morse potential : $V(z)=D_e \left[\left(1 - e^{(-a(z-z_e))} \right)^2 - 1 \right]$

Energy levels : $E_n = 2\pi\hbar\nu \left(n + \frac{1}{2} \right) - \frac{\left[2\pi\hbar\nu \left(n + \frac{1}{2} \right) \right]^2}{4D_e}$ and ZPE : $E_0 = \pi\hbar\nu - \frac{1}{4D_e} \left(\frac{\pi\hbar\nu}{2} \right)^2$

where:
$$\begin{cases} \nu = \frac{a}{2\pi} \sqrt{2D_e/\mu} \\ n = 0, 1, 2 \dots \left[\lambda - \frac{1}{2} \right] \\ \lambda = \frac{\sqrt{2\mu D_e}}{a\hbar} \end{cases}$$

n : vibrational quantum number

ν : state with a unit of frequency

D_e : dissociation energy

μ : reduced mass; $\mu = \frac{m_1 m_2}{m_1 + m_2}$
 $(m_1 = m_{Cu}; m_2 = m_{He} \text{ or } m_{He})$

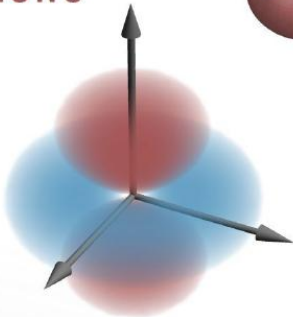
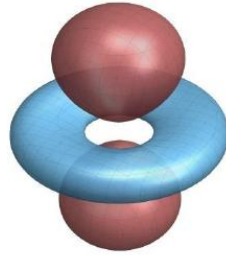
$\hbar$: reduced Planck constant $\left(\frac{h}{2\pi} \right)$

Table 1: Short analysis of the results : At the CCSD(T)/cc-pVTZ level of theory with the Morse parameters ($D_e=5.43 \text{ kJ. mol}^{-1}$, $\alpha = 2.16 \text{ \AA}^{-1}$, $z_e=2.1 \text{ \AA}$)

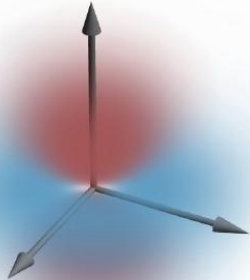
	Analytical results		Numerical results (FDM)		Relative error	
	^4He	^3He	^4He	^3He	^4He	^3He
ZPE kJ/mol	-6.941	-6.839	-6.943	-6.841	0.029	0.029
ΔZPE	0.102		0.102		/	

State of the art

QUANTUM THEORY FOR CHEMICAL APPLICATIONS



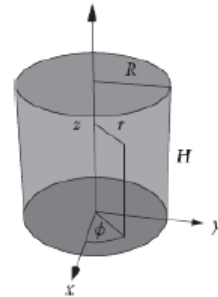
from
basic concepts
to advanced topics



JOCHEN AUTSCHBACH

18.1 Particle in a Cylinder

Many interesting nanoscale systems come in the (approximate) shape of rods. A very simple theoretical model for the quantum effects exhibited by the electrons may confine the electrons to a cylinder shape. This is the Particle in a Cylinder (PiaC) model. The cylinder is defined by its radius R and its height H . Let the axis of the cylinder coincide with the z -axis, and let the linear dimension of the cylinder be $0 \leq z \leq H$, as



$$\hat{H}\psi = -\frac{1}{2}\nabla^2\psi = \psi E \quad \text{with}$$

$$\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2}$$

the normalized radial functions.

$$R_{n_r, m}(r) = \frac{\sqrt{2}}{R J_{m+1}(\alpha_{n_r, m})} J_m\left(\frac{\alpha_{n_r, m}}{R} r\right)$$

The complete normalized wavefunctions for the PiaC are given as

$$\psi_{n_r, m, n_z}(r, \phi, z) = \sqrt{\frac{2}{\pi H}} \frac{1}{R J_{m+1}(\alpha_{n_r, m})} J_m\left(\frac{\alpha_{n_r, m}}{R} r\right) e^{im\phi} \sin\left(\frac{n_z \pi}{H} z\right)$$

with energies

$$E_{r, \phi, z} = \frac{\alpha_{n_r, m}^2}{2R^2} + \frac{n_z^2 \pi^2}{2H^2}$$

Schrödinger equation in 3D using Cylindrical coordinates

The 3D Schrödinger equation's wave equation can be written as follows:

$$\underbrace{\left[-\frac{\hbar^2}{2m^*} \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial z^2} \right) + V(r, \theta, z) \right]}_H \psi(r, \theta, z) = E \psi(r, \theta, z) \quad (1)$$

$$\text{BCs :} \quad \begin{cases} \text{left : } \psi(0, r) = 0 \\ \text{right : } \psi(L, r) = 0 \end{cases}$$

$$\begin{cases} \text{bottom or top : } \psi(R, z) = 0 \\ \text{axial symmetry : } \left. \frac{\partial \psi(z, r)}{\partial r} \right|_{r=0} = 0 \end{cases}$$

✓ Separation of variables

$$\psi(r, \varphi, z) = R(r)Z(z)\phi(\theta); \quad \text{and} \quad V(r, z) = V(r)V(z)V(\theta)$$

$$\text{The laplacian (L) takes the form:} \quad L = \nabla^2 = \frac{1}{r} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial z^2}$$

$$\text{The laplacian (L) takes the form:} \quad r^2 \nabla^2 = r \frac{\partial}{\partial r} + r^2 \frac{\partial^2}{\partial r^2} + \frac{\partial^2}{\partial \theta^2} + r^2 \frac{\partial^2}{\partial z^2}$$

$$\begin{cases} m^*: \text{reduced mass} \\ \hbar: \text{reduced Planck constant} \\ E: \text{energy} \\ \psi(r, z): \text{wavefunctions} \\ V(r, z) = V_M(z) + V(r) + V(\theta) \end{cases}$$

PROBABILITY DENSITY: CARTESIAN COORDINATES

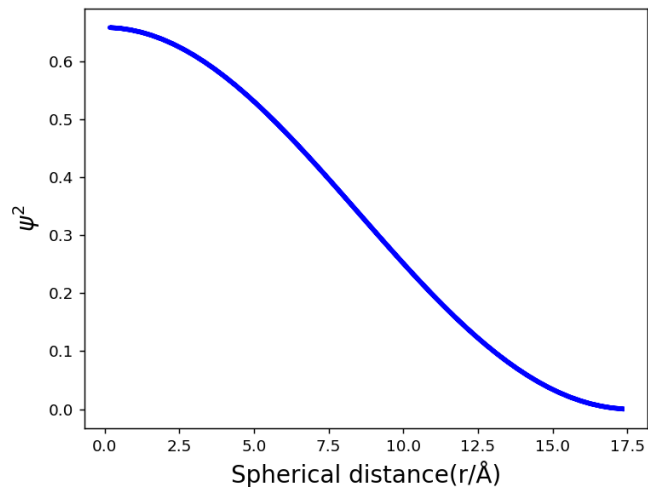


Figure 3: Probability

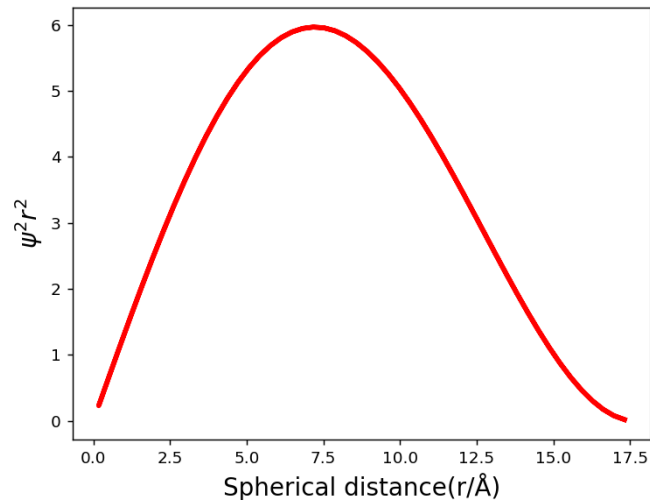
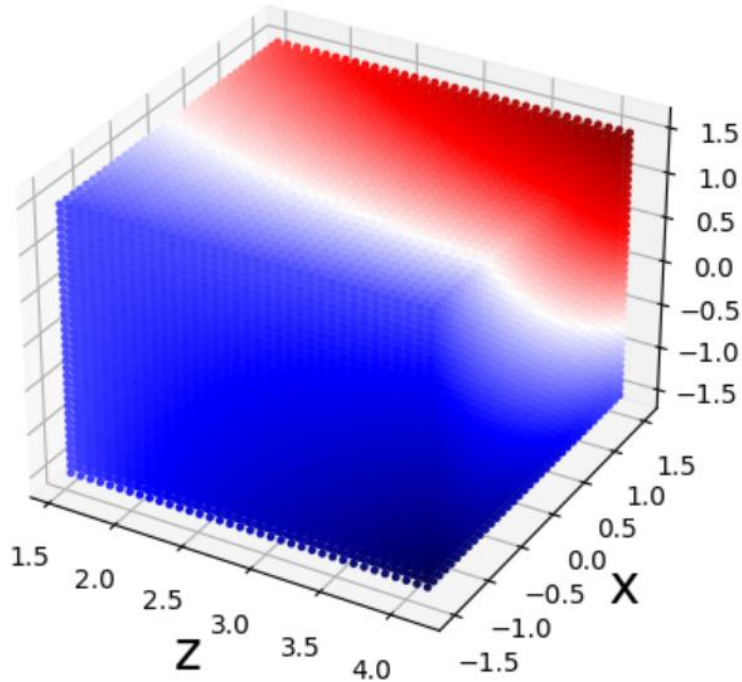


Figure 4: Radial probability

SPARSE MATRICES: CARTESIANS VS CYLINDRICAL COORDINATES.



VALIDATION: $\text{Cu}^+(\text{H}_2\text{O})(\text{H}_2)_2$ SYSTEM

Optimized geometry of $\text{Cu}(\text{H}_2\text{O})(\text{H}_2)_2^+$ system using different approaches

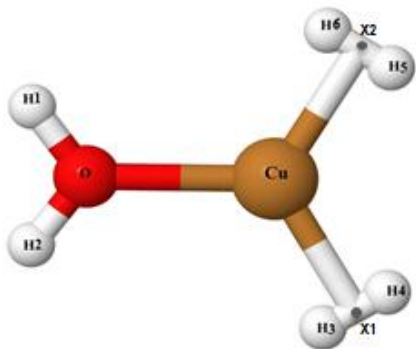


Figure 6. Optimized geometry of $\text{Cu}^+(\text{H}_2\text{O})(\text{H}_2)_2$ at the CCSD(T)/aug-cc-pVTZ(-PP) level and definition of geometry parameters. X1 and X2 represent the centers of mass of H_2 ligands.

- **Table 5.** Geometrical parameters of $\text{Cu}^+(\text{H}_2\text{O})(\text{H}_2)_2$ at different computational levels. Distances in Angstroms ($\text{\AA}$), angles and dihedral angles in degree ($^\circ$), The numbering of the atoms is given in **Figure 6**.

Parameters	MP2/ def2- TZVPP	MP2/ aug-cc- pVTZ - PP	CCSD(T)/ aug-cc-pVTZ- PP
$r(\text{Cu-O})$	1.947	1.928	1.948
$r(\text{Cu-X1/X2})$	1.651	1.587	1.642
$r(\text{O-H1/H2})$	0.963	0.965	0.964
$r(\text{H3-H4})$	0.789	0.789	0.790
$\angle(\text{H1-O-H2})$	106.93	107.23	107.15
$\angle(\text{H2-O-Cu})$	126.53	126.38	126.43
$\angle(\text{X1/X2-Cu-O})$	119.98	119.52	120.64